

# Numerical methods

## Introduction

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**Course:** Numerical methods – Lecture  
Computer Science BSc

**Code:** IP-18fNM1E

**Semester:** 2023/2024 Spring

**Location:** South Building, Room 0-804 (L. Lóczy)

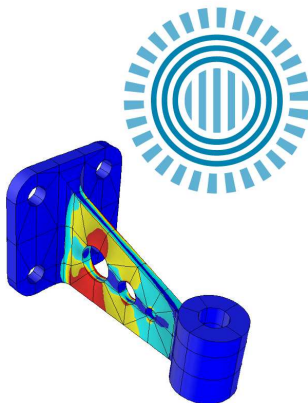
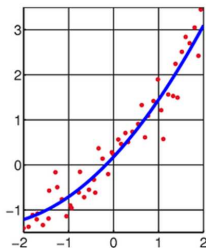
**Time:** Tuesdays 12:15 – 13:45

# What is this course about?

The aim of numerical analysis is to elaborate and analyse methods for the exact or approximate **computer aided** solution of problems in mathematics, **engineering and sciences**.

This is a one semester course in the new (2018) curriculum discussing the numerical methods of **linear algebra** and **analysis**.  
An introductory course to numerical methods.

$$\sqrt{2}$$



weather forecast

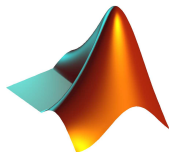
sources:  $\text{\LaTeX}$ , Prezi, WinAmp, Wikipedia

- Machine numbers. Error analysis.
- Direct methods for the solution of linear systems of equations.  
(Gaussian elimination, LU decomposition)
- Matrix norms. Perturbations of linear systems.
- Estimating roots of polynomials. Horner's algorithm.
- Solving non-linear equations.  
(bisection method, fixed point iteration, Newton's method, secant method, false position)
- Interpolation.
- Least squares methods, approximation.
- Quadrature, numerical integration.

- **Presentation slides in Canvas.**
- Recordings of an earlier year's presentations shared
- Csaba Hegedűs: Numerical methods I. and II.
- Written notes (problems & solutions) for the practice.

## ① Some sample problems, examples in Matlab.

(Available for ELTE students. Not required, you're welcome to try.)



## ② Presentations.

(See Canvas. Definitions, theorems, proofs, sample problems, additional notes.)

## ③ Interactive discussion.

Since all material is available in detail.



## ① Practical mark

- tests
- homework assignments (for extra points)
- cf. practice

## ② Exam

- **following** the lectures is compulsory. 3 misses allowed.
- **written part:** must score at least 8 points out of 15 (45 minutes). Sample will be provided. For grades *pass* (2) and *satisfactory* (3). Below 8 points: *fail* (1).
- **oral part:** If written part is passed. A more in depth discussion on a randomly chosen topic. For grades *good* (4) and *excellent* (5). Both part's results are considered for the final grade.

Fill the *attendance quiz* in Canvas in the given time frame appropriately!  
Both for lecture and practice (for Neptun group).



- Describe an aspect of reality (physics, chemistry etc.) via a *mathematical model* to perform calculations. This may only be an approximation: **model error**.
- The exact solution usually can not be found in finite steps (instead of arbitrary 'infinite' precision), *approximation methods* are needed for the computer program: **approximation error**.

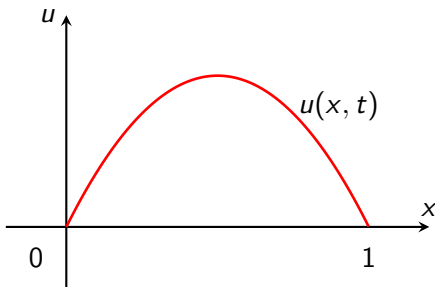
- The *input parameters* of the model are usually measured values (with limited accuracy): **measurement (or inherited) error**.
- We put the input in finite computer arithmetics: **input error**.
- Calculation in finite arithmetics may lead to overflow, underflow, or rounding errors: **arithmetic (round off) error**.
- The implemented approximation method is tested, *verified* against the expected results.  
If the results are not sufficiently accurate, we start again with refining the above steps.

## An example: the oscillating string

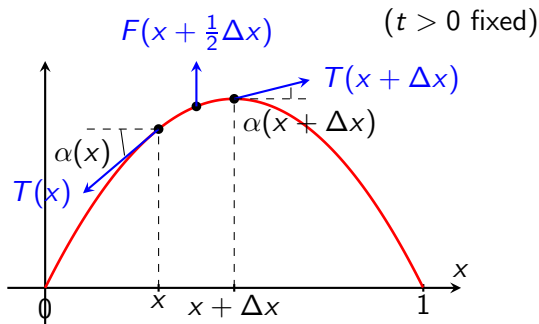
Expand a string with fixed endpoints and let go ( $t = 0$ ).  
Let it oscillate.

**Task:** describe the oscillation:

find  $u(x, t)$  positions at point  $x$  at time  $t$ .



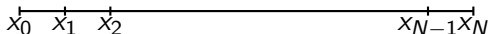
# An example: the oscillating string



- Physical assumptions...
- Leads to a partial differential equation...

# An example: the oscillating string

- Discretisation, finite points...



- We arrive at a linear system of equations (in a simplified case).

$$\begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ \ddots & \ddots & \ddots & & \vdots \\ 0 & \dots & 0 & -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ \vdots \\ u_{N-1} \end{bmatrix} = -h^2 \cdot \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ \vdots \\ A_{N-1} \end{bmatrix}$$

$(N-1) \times (N-1)$  linear system. A tridiagonal system which may be solved using (quick) Gaussian elimination.